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Why Core GRADE is needed: introduction to a new series in *The BMJ*

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This article introduces a series of papers on new guidance for the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach. Core GRADE was produced in response to the need for a concise, well organised exposition of the key elements of GRADE that users require to make optimal judgments about certainty of evidence and strength of recommendations. This series is primarily aimed at systematic review authors, guideline developers, and health technology assessment practitioners, along with evidence based medicine educators who help clinicians to understand and use GRADE to guide clinical care. In producing Core GRADE, the authors address the problems of out-of-date and poorly organised guidance resulting from publication of the many GRADE papers over 20 years.

How evidence is synthesised, made accessible, and used in guidelines has improved greatly in the past 40 years.[1] One such improvement was the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach to rating certainty of evidence and moving from evidence to recommendations. When one of us (GG) proposed the concept of evidence based medicine (EBM) in the early 1990s and introduced guides on its use,[2] it was as a way to teach and learn how to make use of clinical research in decision making in patient care. The EBM approach democratised the decision making process and made it more widely accessible. With a focus on improving accessibility further, we produced Core GRADE—new guidance comprising the essential elements of GRADE.

One limitation that has arisen from improvements in the processing of evidence is that guidance for assessing and synthesising evidence, developing guidelines, and informing clinical decisions and health policy has become more complex—sometimes too complex.[3-6] Examples include assessing risk of bias in studies, preparing systematic reviews, and developing guidelines.[3 4 7]

Since its introduction in 2004, the GRADE approach to rating certainty (quality) of evidence in systematic reviews and grading strength of recommendations[8] has had an important effect on the conduct of systematic reviews and on clinical practice guidelines. All systematic reviews performed by the Cochrane Collaboration are required to follow the GRADE approach. GRADE is also endorsed and used by more than 120 organisations worldwide, including the World Health Organization (WHO), American College of Physicians, the online resource UpToDate, and *The BMJ*. Although initially directed at clinicians providing care to individual patients, GRADE now applies to other specialties, such as public health and environmental health.[9] The GRADE approach has also extended its coverage from issues of patient management to guidance on diagnosis,[10] prognosis,[11] and qualitative research.[12]

Reasons for the widespread uptake of the GRADE approach include its practicality and user friendliness, combined with rigour, sophisticated methodology, and availability of detailed guidance for authors of systematic reviews and developers of guidelines and health technology assessment reports. More than 50 publications now provide guidance on GRADE for these key target audiences, as well as for clinician users.

As a result of the expansion and continuous development of more detailed GRADE guidance, however, fundamental challenges have become apparent. Firstly, GRADE has become increasingly complex, such that new as well as existing users may find it unwieldy. Secondly, keeping up with evolving GRADE guidance has proved daunting for users required to find and process continually appearing new guidance that is poorly organised and difficult to navigate.[13-16]

SUMMARY POINTS

- The GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach is the dominant system of rating certainty of evidence and grading healthcare recommendations worldwide
- Since its introduction in 2004, the GRADE approach had become increasingly complex and now requires reformulation for optimal use
- Core GRADE was developed to provide the essentials of the GRADE approach in a single series of papers
- Core GRADE focuses on the comparison of a single intervention with a single comparator in the context of clinical decision making
- Although largely consistent with previous GRADE papers, Core GRADE presents key advances that are now part of the essential approach

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The papers in this new series were produced in response to the need for a concise, well organised exposition of the key GRADE elements that users require to ensure that their judgments about certainty of evidence and strength of recommendations are optimal. This series is mainly aimed at systematic review authors, guideline developers, and health technology assessment practitioners—individuals who compare clinical management strategies and are interested in using GRADE. Another audience is EBM educators who help clinicians to understand and use GRADE to guide clinical care. Core GRADE was produced to solve the problem of outdated and poorly organised guidance resulting from the sequential publication of the many GRADE papers over 20 years.

Core GRADE focuses on the comparison between a single intervention and a single comparator in the context of clinical decision making. Several articles in a series published in the *Journal of Clinical Epidemiology* from 2011[17] addressed problems relevant to this comparison: Core GRADE represents a rationalisation, reformulation, extension, and modification of GRADE principles of rating certainty of evidence and moving from evidence to recommendations. Although largely consistent with previous GRADE papers, Core GRADE presents key advances that are now part of the essential approach. Core GRADE overcomes the problem of complexity by including key guidance necessary to make sound judgments on the trustworthiness of evidence and the appropriate direction and strength of recommendations. The Core GRADE series is comprehensive: by presenting all the concepts and approaches required to implement Core GRADE, users no longer need to seek out guidance from previous GRADE papers.

Fundamental principles of Core GRADE include its flexibility: more than one best judgment is often available for certainty of evidence and moving from evidence to recommendations. Thus, we offer guidance, not rules. The recognition that there are often close-call decisions when rating certainty of evidence further reinforces the principle of flexibility. Other key principles include practicality and simplification of concepts.

We also developed principles for what to leave out of Core GRADE. For example, when an incremental gain would be questionable, we have not included more complex guidance. Criteria for identifying such issues are that they have been in the public domain for several years with little uptake, they are sufficiently arcane to deter many review and guideline producers from learning about them, they change often, and they seldom arise as problems that authors confront.

The ultimate criterion, however, dictates that if a guidance is not necessary to make sound judgments on certainty of evidence and direction and strength of recommendations, it is not part of Core GRADE. This is the case even when the guidance contributes elegance, depth of understanding, enhanced transparency, or minor modifications of judgments on certainty of evidence. This stringent standard defines the concept of rigorous simplicity we adopted for Core GRADE and ensures its optimal accessibility, usefulness, and dissemination.

To judge these principles, the authors relied on their extensive experience of applying GRADE. As a group we have provided supervisory support for hundreds of systematic reviews, most with lead authors who were students or others with limited expertise: the audience for Core GRADE. We also have experience of hundreds of clinical guidelines and health technology assessment reports produced by a wide variety of groups, most of whom—including WHO guideline groups and medical specialty organisations—recruit panels in which most of the members have a limited GRADE background.

Our experience in teaching GRADE to students and guideline panel members for more than 20 years, and the resulting insights on their needs in applying GRADE, also proved of great value. Extensive and detailed discussion and an ultimate consensus among the authors guided the application of the principles of what optimally belongs in Core GRADE.

This new series introducing Core GRADE comprises seven papers. The first paper presents an overview of Core GRADE dealing with certainty of evidence and moving from evidence to recommendations or decisions. The second paper addresses key issues in rating certainty of evidence: the target of certainty rating, study design, and rating down for imprecision. The third paper addresses issues of inconsistency: large differences in results from study to study. The fourth paper addresses risk of bias, the possibility of rating up certainty for large effects and dose-response gradients, and concerns of publication bias. The fifth paper addresses concerns related to indirectness (issues of applicability and surrogate outcomes). The sixth paper addresses the creation of Core GRADE summary of findings tables, the key single source that presents the effects of interventions on outcomes of importance to patients. The seventh and final paper addresses issues that guideline developers and health technology assessment practitioners face when moving from evidence to recommendations or decisions.

Core GRADE should help to overcome the problems with current GRADE guidance that compromise its effective and optimal use. Its availability should enable the approach to achieve its full potential in optimising clinical decision making through providing trustworthy judgments of evidence certainty in systematic reviews and health technology assessment guidance, and recommendations in clinical practice guidelines.

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Core GRADE comes not as an official product of the GRADE Working Group, but rather as the collaborative effort of a group of GRADE experts, building on their pivotal roles in much of previous GRADE guidance, or applying GRADE through the MAGIC Evidence Ecosystem Foundation, developing BMJ Rapid Recommendations,[18] and innumerable systematic reviews and guidelines.

Contributors: GG, VMM, TA, MH, and AI conceived and designed the Core GRADE series and are guarantors for the paper. GG and VMM drafted the manuscript. All authors critically revised the manuscript across several iterations for important intellectual content and gave final approval of the article. The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

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